Skills Summary

Diagnosing Transformation-Order and Inverse-Domain Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Inside the function or outside it

Outside f ($a\,f(x)+k$): stretches and shifts the *output*, vertically, exactly as written.

Inside f ($f(bx+c)$): changes the *input*, horizontally, and always in the *opposite* sense — $f(x+3)$ moves left, $f(2x)$ compresses.

Example: $f(x) = x^2$. Evaluate $g(x) = f(x+3)$ at $x = 1$.
Step 1. The $+3$ sits inside f, so it changes the input before squaring — it is not added to the answer.
Step 2. $g(1) = f(1+3) = f(4) = 4^2$.
$\Rightarrow$ $\boldsymbol{g(1) = 16}$

Try it: Same f. Evaluate $h(x) = f(x) + 3$ at $x = 1$, and notice how far apart the two answers are.

check: $\boldsymbol{h(1) = 4}$

Skill 2 — Ordering the outside operations

Rule. A vertical stretch multiplies everything already present, so *stretch and reflect first, shift last.*

Written out: $a\,f(x)+k$ means take $f(x)$, multiply by a, *then* add k. Doing the shift first turns it into $a\big(f(x)+k\big)$, which moves the graph by ak instead of k.

Example: Stretch $y = |x|$ vertically by 4, then shift down 5. Find $g(3)$.
Step 1. The shift is last, so it must stay outside the multiplication — writing a bracket around both would multiply the shift too.
Step 2. $g(x) = 4|x| - 5$, so $g(3) = 4(3) - 5 = 12 - 5$.
$\Rightarrow$ $\boldsymbol{g(3) = 7}$

Try it: Stretch $y = x^2$ vertically by 2, then shift up 3. Find $g(4)$.

check: $\boldsymbol{g(4) = 35}$

(!) Watch out: a reflection is a stretch by a negative number, so it obeys the same rule — reflect before shifting, or the shift gets negated too.

Skill 3 — Letting the restriction pick the branch

Rule. A squared rule sends two inputs to one output, so it has an inverse only on one side of its vertex.

The kept side decides the sign: a domain *above* the vertex gives $h + \sqrt{\cdots}$, a domain *below* it gives $h - \sqrt{\cdots}$. Check by asking whether your answer lands inside the original domain.

Example: $f(x) = (x-1)^2 + 2$ on $x \geq 1$. Find $f^{-1}(11)$.
Step 1. The domain starts at the vertex and runs right, so the inverse must return values at or above 1 — that fixes the plus sign before any arithmetic.
Step 2. $f^{-1}(x) = 1 + \sqrt{x-2}$, so $f^{-1}(11) = 1 + \sqrt{9} = 1 + 3$.
$\Rightarrow$ $\boldsymbol{f^{-1}(11) = 4}$

Try it: $f(x) = (x+2)^2 - 1$ on $x \leq -2$. Find $f^{-1}(24)$.

check: $\boldsymbol{f^{-1}(24) = -7}$